

Mode Choice

Discrete choice and the binary logit model

Lewis Lehe

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Recap: the four-step model

1. Trip Generation

How many trips does each zone produce and attract?

2. Trip Distribution

*Where do trips from zone i go?
(origin-destination matrix)*

3. Mode Choice

**Which mode do travelers use?
(today's lecture)**

4. Traffic Assignment

*Which routes are taken?
(link flows in the network)*

Purpose of the mode choice stage

- **Input:** an origin-destination (O-D) matrix T_{ij} from trip distribution, total trips from zone i to zone j .
- **Output:** an O-D matrix *for each mode*, T_{ij}^m , where $m \in \{\text{auto, transit, walk, bike, ...}\}$.
- Equivalently: predict the **share** of travelers in each (i, j) pair who use each mode.
- Why we care:
 - Forecasting transit ridership (will a new line attract enough riders?)
 - Estimating vehicle emissions (mode share $\rightarrow$ vehicle-miles)
 - Evaluating policy: gas taxes, parking fees, free-fare transit

What we need from a mode-choice model

- Travelers do not all make the same choice, otherwise everyone with the same options would pick the same mode.
- So the model must produce a **probability** for each mode, not a single answer.
- We want those probabilities to depend on observable attributes:
 - travel time by each mode
 - cost (fare, fuel, parking)
 - number of transfers, walk distance, etc.
- And on traveler characteristics (income, household size, vehicle ownership).

BART Example

Daniel McFadden and the BART forecast

- In the late 1960s, the San Francisco Bay Area was building **BART**: a brand new heavy-rail commuter system.
- **Daniel McFadden**, an economist at UC Berkeley, estimated a discrete-choice model of mode share for the journey-to-work, using survey data of commuters *before* BART existed.
- His pre-opening forecast: BART would carry about **6.3%** of commute trips.
- After BART opened, the actual share was **6.2%**, well within the model's predicted standard error.
- McFadden won the **Nobel Prize in Economics in 2000** for this line of work.

Prediction success table (Journey-to-Work)

Pre-BART model vs. post-BART choices (cell counts)

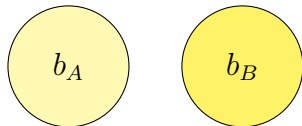
	Auto Alone	Carpool	Bus	BART	Total
Auto Alone (actual)	255.1	79.1	28.5	15.2	378
Carpool (actual)	74.7	37.7	15.7	8.9	137
Bus (actual)	12.8	16.5	42.9	4.7	77
BART (actual)	9.8	11.1	6.9	11.2	39
Total	352.4	144.5	94.0	40.0	631
Predicted share	55.8%	22.9%	14.9%	6.3%	
(Std. error)	(11.4%)	(10.7%)	(3.7%)	(2.5%)	
Actual share	59.9%	21.7%	12.2%	6.2%	

- Diagonal cells = correct predictions. Off-diagonal = predicted-actual mismatches.
- BART share predicted **6.3%**, actual **6.2%**, within $\pm 2.5\%$ standard error.
- How does a model that has *never seen* BART predict its share so well?

Where does the math come from?

Psychophysics: which light is brighter?

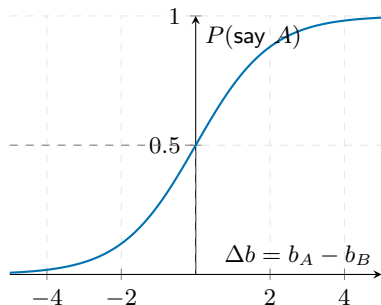
- Classic 19th-century experiment. Show a subject two lights:
 - Light A with brightness b_A
 - Light B with brightness b_B
- Ask: “which one is brighter?”
- Vary the difference $\Delta b = b_A - b_B$ across trials.
- Record the fraction of times the subject says “ A ”.
- The brain's perception of brightness has **noise**. Two lights of the *same* brightness do not always feel the same.



which is brighter?

The sigmoid emerges

- Plot $P(\text{subject says } A)$ as a function of $\Delta b = b_A - b_B$.
- The curve is an **S-shape (sigmoid)**:
 - $P = 50\%$ when $\Delta b = 0$
 - $P \rightarrow 100\%$ as $\Delta b \rightarrow +\infty$
 - $P \rightarrow 0\%$ as $\Delta b \rightarrow -\infty$
 - symmetric around 0
- This is the **same shape** we will get for travel-mode choice.



From light to utility

- Now swap the lights for travel modes. Each option i has a **systematic utility** V_i :

V_i = what the analyst can compute from observable attributes.

- For travel: V_i depends on travel time, cost, transfers, comfort, etc.
- This is the analog of each light's *true* brightness.
- But just as the brain's brightness perception is noisy, real travelers respond to many things we do *not* observe: mood, weather, schedule quirks, taste.
- So the actual utility a traveler experiences is

$$U_i = V_i + \varepsilon_i,$$

where ε_i is the unobserved (random) part.

The random utility model

- Each traveler picks the option with the **highest** U_i :

$$\text{chosen mode} = \arg \max_i (V_i + \varepsilon_i).$$

- Because ε_i is random, the analyst cannot predict the choice with certainty, only its **probability**:

$$P_i = \Pr[V_i + \varepsilon_i > V_j + \varepsilon_j \text{ for all } j \neq i].$$

- The shape of P_i depends on how ε is distributed.
- If ε has a **Gumbel** (extreme-value) distribution, the resulting P_i is exactly the sigmoid we plotted: the **logit model**.

Example: car vs bus

A student's mode choice

- A student picks between **driving** and **taking the bus** to school.
- Bus is **free** (universal student pass).
- Numbers (illustrative):
 - Car: $T_{\text{car}} = 15 \text{ min}$, $C_{\text{car}} = \$3$ (gas + parking)
 - Bus: $T_{\text{bus}} = 30 \text{ min}$, $C_{\text{bus}} = \$0$
- Estimated coefficients:
 - $\alpha_{\text{car}} = 0.5$ (mode-specific constant)
 - $\beta_T = -0.05$ per minute
 - $\beta_C = -0.30$ per dollar

Drive

15 min
\$3

Bus

30 min
free

Computing the systematic utilities

- Utility specification:

$$V_{\text{car}} = \alpha_{\text{car}} + \beta_T T_{\text{car}} + \beta_C C_{\text{car}}, \quad V_{\text{bus}} = \beta_T T_{\text{bus}} + \beta_C C_{\text{bus}}.$$

- Plug in:

$$V_{\text{car}} = 0.5 + (-0.05)(15) + (-0.30)(3) = 0.5 - 0.75 - 0.90 = -1.15$$

$$V_{\text{bus}} = 0 + (-0.05)(30) + (-0.30)(0) = -1.50$$

- Notice the trade-off: bus loses on time (15 extra minutes) but wins on cost (\$3 cheaper). $V_{\text{car}} > V_{\text{bus}}$ by 0.35 utils.
- Does that mean every student drives? **No**, because of ε .

The binary logit formula

- With Gumbel-distributed ε , the random-utility argmax simplifies to

$$P(A) = \frac{e^{V_A}}{e^{V_A} + e^{V_B}}$$

- Equivalently, in terms of the utility difference $\Delta V = V_A - V_B$,

$$P(A) = \frac{1}{1 + e^{-\Delta V}}.$$

- This is the **same sigmoid** we drew for the brightness experiment, with ΔV in place of Δb .
- The denominator just normalizes so that $P(A) + P(B) = 1$.

Plug in the numbers

- Exponentiate the systematic utilities:

$$e^{V_{\text{car}}} = e^{-1.15} \approx 0.317, \quad e^{V_{\text{bus}}} = e^{-1.50} \approx 0.223.$$

- Probabilities:

$$P(\text{car}) = \frac{0.317}{0.317 + 0.223} = \frac{0.317}{0.540} \approx \mathbf{58.7\%}$$

$$P(\text{bus}) = \frac{0.223}{0.540} \approx \mathbf{41.3\%}$$

- So even though bus is slower, the cost advantage keeps a substantial share of students on the bus.
- And this is for a single (T, C) point, across a whole population, the model gives a continuous distribution of mode shares.

What changes the choice?

- The model lets us answer “what if” questions just by recomputing V .
- Suppose the city builds a bus lane and bus time drops from 30 to **20 minutes**:

$$V_{\text{bus}}^{\text{new}} = (-0.05)(20) + 0 = -1.00$$

$$e^{V_{\text{bus}}^{\text{new}}} \approx 0.368$$

$$P(\text{bus}) = \frac{0.368}{0.368 + 0.317} \approx \mathbf{53.7\%}$$

- Bus share rises from 41% to 54%, enough to flip the majority.
- More generally: shrinking $|V_{\text{car}} - V_{\text{bus}}|$ pushes shares toward 50/50; growing it pushes them toward 0/100.

Where do the α 's and β 's come from?

- The coefficients are not picked by hand. They are **estimated** from data on real choices.
- Two main kinds of choice data:
 - **Revealed preference (RP)**: what people actually did. Travel diaries, transit smart-card swipes, household travel surveys.
 - **Stated preference (SP)**: hypothetical questions in a survey. "If a new bus had 15-minute headways and a \$2 fare, would you ride it?"
- **RP** is real behavior, but limited to options that already exist (you can't observe BART trips before BART opens).
- **SP** can probe brand-new alternatives, but answers may not match what people would do for real.
- Practitioners often **combine** both.
- Given the data, the coefficients are fit by **maximum likelihood**: pick the α, β that make the observed choices most probable.

What we did not do

- **More than two modes.** With K modes, the binary formula generalizes to the **multinomial logit**:

$$P_i = \frac{e^{V_i}}{\sum_{k=1}^K e^{V_k}}.$$

- **Model limitations.** The logit's "Independence of Irrelevant Alternatives" assumption can break (the *red bus / blue bus* paradox). Fixed by nested or mixed logit.
- These are big topics. McFadden and others won prizes for them.

Recap

- **Mode choice** is stage 3 of the four-step model: split each O-D flow T_{ij} into mode-specific flows T_{ij}^m .
- **Random utility model**: each option has a systematic utility V_i from observables, plus a random ε_i for unobservables. Travelers pick $\arg \max_i (V_i + \varepsilon_i)$.
- Borrowing from **psychophysics**: the choice probability is a sigmoid in the utility difference, exactly like “which light is brighter?”.
- **Binary logit**:

$$P(A) = \frac{e^{V_A}}{e^{V_A} + e^{V_B}}.$$

- **McFadden + BART** (1972): a discrete-choice model fit *before* BART opened predicted its commute share within 0.1%.
- **Next time**: stage 4, traffic assignment.